Michael Capriotti

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Education

Northwestern University - Bachelor's degree in **Computer Science and Mathematics**, **GPA**: 3.92/4.0 — *Evanston, IL*

- **Relevant Coursework:** MENU Linear Algebra & Multivariable Calculus, Data Structures & Algorithms, Operating Systems, Design & Analysis of Algorithms, Discrete Mathematics, Human-Computer Interaction — *2024 - 2028*

Relevant Work Experience

NASA Marshall Space Flight Center — Huntsville, AL — *June 2026 - August 2026*

Software Engineering Intern

- Developed a PostgreSQL database comprising 40+ normalized tables for mass properties data, replacing a fragmented, manual spreadsheet-based workflow; supported hierarchical vehicle data, revision tracking, and reusable item records while preserving an immutable history; optimized queries with targeted indexes and enforced data integrity through constraints and triggers.
- Migrated 4M+ historical mass property values into the database using Python data-processing pipelines and a standardized file format that isolates vendor-specific parsing from database logic, enabling new vendor formats without modifying the core pipeline.
- Built a FastAPI application for mass properties workflows, including search, hierarchy navigation, revision management, and import/export; developed 200+ unit and end-to-end tests covering operations and processing workflows, achieving $\sim$ 90% coverage.

Northwestern Computer Science Department — Evanston, IL — *January 2026 - June 2026*

Undergraduate Teaching Assistant

- Supported a computer systems course with 150+ students through tutoring, exam proctoring, grading, and instruction on data representation, assembly, the stack, buffer overflow attacks, virtual memory, caches, file systems, concurrency, and process scheduling.

Research Projects

Quantum Computing Research at Los Alamos National Laboratory — *Python, NumPy, SciPy, Matplotlib, Qiskit, CVXPY*

- Applied the Quantum Approximate Optimization Algorithm (QAOA) to solve combinatorial optimization problems (portfolio optimization, traveling salesman, and maximum independent set) by transforming QUBO problems into Max-Cut instances.
- Utilized semidefinite programming (SDP)-based warm-start strategies to leverage classical optimization for improved QAOA initialization and faster convergence. Submitted a revised manuscript to ACM Transactions on Quantum Computing.

Data Science Research at Northwestern Kellogg School of Management — *Python, Pytesseract, OCR, LLMs*

- Developed a scalable automated obituary classification pipeline to assess executive importance, transitioning from manual labeling (500 samples) to OCR extraction and LLM-based labeling. Presented work at IMSAloquium.

Software Projects

Stock Forecast — *Python, TensorFlow, Pandas, FastAPI, React, TypeScript, SQLite*

- Built a machine learning web application predicting next-day closing prices for 400+ companies using LSTM models trained on 5 years of historical market data, achieving an average mean absolute error (MAE) of 2.76%, with interactive visualizations.

Golf Swing Analyzer — *Python, Scikit-learn, Random Forest, MediaPipe, OpenCV, Flask*

- Engineered and deployed a full-stack web application for golf swing analysis, processing 100 labeled swings and extracting normalized 3D coordinates from 13 key body landmarks to classify videos as Pro or Amateur with a Random Forest model achieving $\sim$ 80% accuracy; enabled video upload, trimming, and interactive pose annotations while minimizing backend memory usage.

Trading Simulation — *C++, STL*

- Created a single-ticker market simulator with a price-sorted order book and matching engine, implementing four trading strategies (Value Investor, Trend Follower, Stop Loss, Market Maker) and simulating execution and portfolio performance over 2,000 ticks.

Evolutionary Flappy Bird Game — *C#, Unity, Evolutionary Algorithm*

- Implemented a custom NEAT-inspired neural network in C# with elitism, mutation, and multi-cycle fitness evaluation, enabling progressive AI improvement across generations and automatic preservation of the best-performing models for replay.

Donation Tracking Platform — *JavaScript, Node.js, PostgreSQL, AWS (Lambda, RDS), Firebase*

- Developed the database and backend APIs for a donation tracking platform used by a nonprofit through Northwestern DISC, enabling secure role-based access, Excel data uploads, validation, filtering, and donation history queries for a frontend dashboard.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, C#, SQL, HTML/CSS.

Frameworks/Platforms: FastAPI, React, Node.js, Flask, AWS, Docker, GitHub Actions, Unity.

Libraries: NumPy, Pandas, SciPy, CVXPY, Scikit-learn, TensorFlow, Qiskit, Matplotlib, MediaPipe, Pytesseract

Additional Information

Programs: MIT Introduction to Technology, Engineering, and Science (MITES).

Activities: Develop & Innovate for Social Change (DISC), Consultants Advising Student Enterprises, Table Tennis Club.